

Do Gold and Bitcoin Expand the Opportunity Set for Australian Superannuation Members?

Empirical implementation of the Assessment 1 proposal. This repository collects the data, builds the monthly return panel, and produces the descriptive statistics, tables and figures that the mean–variance spanning tests (Huberman–Kandel 1987; Kan–Zhou 2012) and the out-of-sample portfolio tests will build on.

The repository contains **code and documentation only**. No data files are distributed. Running the pipeline recreates every dataset from its original source and regenerates every output.

Quick start

```
# 1. Open Assignment2.Rproj in RStudio (this sets the working directory)
# 2. Copy .Renviro.example to .Renviro and fill in the two values,
#    then restart R so they are loaded
# 3. Install dependencies, once:
source("R/install_packages.R")
# 4. Run the whole pipeline:
source("R/00_run_all.R")
```

Roughly 3–5 minutes end to end. Tables appear in `output/tables/`, figures in `output/figures/`.

Before running, read [Data sources and access requirements](#). Two of the seven series need more than an internet connection.

Research design implemented here

Role	Series
Benchmark assets	DIY options of AustralianSuper and ART , analysed separately: Australian shares, international shares, fixed income
Test asset 1	Gold — CRSP gold ETF in USD converted to AUD at the RBA rate (primary); ASX-listed gold ETF in AUD (alternative)
Test asset 2	Bitcoin — BTC-AUD
Risk-free rate	RBA cash rate target (table F1.1, <code>FIRMMCRT</code>) — reported only, see below

All statistics are computed on **monthly total returns**, which is the quantity the spanning tests use. This follows the classic Huberman–Kandel (1987) setup: the two restrictions $\alpha = 0$ and $\delta = 1 - \beta'1 = 0$ are tested on raw returns, and the δ restriction exists precisely because no risk-free asset is assumed.

The cash rate is still collected, but it enters only the `Sharpe (vs cash)` column of the descriptive tables, where a ratio of mean return to standard deviation would otherwise not be a Sharpe ratio. It is retained because the Stage 2 performance measures (Sharpe, Sortino, Omega) require it.

Two sample windows, as specified in the proposal:

Sample	Window	Months	Contents
Full	2008-08 → 2024-12	197	Benchmarks and gold; Bitcoin does not yet exist
Matched	2014-11 → 2024-12	122	Benchmarks, gold and Bitcoin over identical months

Bitcoin's start sets the beginning of the matched sample. The end of the CRSP licence, not the data itself, sets the end of both.

Repository structure

```

Assignment2/
├── Assignment2.Rproj      RStudio project; sets the working directory
├── README.md             this file
├── .gitignore            excludes data and credentials
├── .Renviron.example     template for the two local credentials
├── R/
│   ├── 00_run_all.R      runs steps 01-04 in order
│   ├── 01_collect_data.R original sources    → data/raw/
│   ├── 02_clean_merge.R data/raw/        → data/processed/
│   ├── 03_analysis.R    panel            → descriptive results
│   ├── 04_tables_figures.R results        → output/
│   └── install_packages.R dependency installer
├── data/
│   ├── raw/              sources exactly as downloaded
│   └── processed/        cleaned series, monthly panel, results
├── output/
│   ├── tables/           tab01-tab06 (.csv)
│   └── figures/          fig01-fig06 (.png)
└── docs/
    ├── data_sources.md   detailed source and access notes
    └── session_info.txt  R and package versions of the last full run

```

Raw data is never overwritten. `01_collect_data.R` writes only to `data/raw/`, and every filtering, conversion and column decision happens in `02_clean_merge.R`. Any cleaning choice can therefore be revisited without downloading again.

Outputs

Tables

File	Contents
<code>tab01_data_sources.csv</code>	Series, source, access requirement, coverage obtained
<code>tab02_descriptives_full_sample.csv</code>	Descriptive statistics, full sample
<code>tab03_descriptives_matched_sample.csv</code>	Descriptive statistics, matched sample
<code>tab04_correlation_full_sample.csv</code>	Correlation matrix, full sample
<code>tab05_correlation_matched_sample.csv</code>	Correlation matrix, matched sample
<code>tab06_test_vs_benchmark_correlation.csv</code>	Each test asset against each fund's options

Descriptive tables report N, mean, median, SD, min, max, skewness, excess kurtosis, a Jarque–Bera p-value, AR(1), annualised mean and SD, all on total returns, plus a Sharpe ratio computed against the cash rate. AR(1) is included because the proposal uses Newey–West HAC standard errors, which matter when returns are autocorrelated.

Figures

File	Contents
<code>fig01_cumulative_growth.png</code>	Cumulative growth of A\$100, total returns, log scale
<code>fig02_risk_return.png</code>	Annualised risk against total return, both samples, log-log
<code>fig03_correlation_heatmap.png</code>	Correlation matrix, matched sample
<code>fig04_rolling_correlation.png</code>	24-month rolling correlation with each fund's Australian shares
<code>fig05_return_distributions.png</code>	Distribution of monthly total returns
<code>fig06_data_coverage.png</code>	Coverage of each series and the matched sample window

Data sources and access requirements

#	Source	Series	Access	Requirement
1	AustralianSuper	3 DIY options	Internal API behind a bot filter	Chrome or Edge must be installed
2	Australian Retirement Trust	3 DIY options	Undocumented internal API	<code>ART_SUBSCRIPTION_KEY</code> in <code>.Renvirom</code> ; see <code>.Renvirom.example</code>
3	WRDS / CRSP	Gold ETF (GLD) in USD	PostgreSQL	Licensed WRDS account required
4	Yahoo Finance	ASX gold ETF (GOLD.AX)	Unofficial public API	None; no service agreement
5	Yahoo Finance	Bitcoin (BTC-AUD)	Unofficial public API	None
6	RBA table F11	AUD/USD	Direct file download	None
7	RBA table F1.1	Cash rate (<code>FIRMMCRT</code>)	Direct file download	None

Full detail, including endpoints, required headers and the reverse-engineering notes for sources 1 and 2, is in `docs/data_sources.md`.

AustralianSuper needs a browser

The site is behind an Akamai bot filter that returns HTTP 403 to every non-browser client. Confirmed against `httr2` with full browser headers and `download.file()` on both the `wininet` and `libcurl` backends. The filter fingerprints the TLS client, so no combination of headers helps.

`01_collect_data.R` tries a direct request first, then falls back to a headless Chrome or Edge. **Browser paths are currently Windows-specific.** On macOS or Linux set the location first:

```
Sys.setenv(CHROMOTE_CHROME = "/path/to/chrome")
```

WRDS needs a licence

Requires an institutional WRDS account with a CRSP subscription. Authentication uses the standard WRDS `pgpass` file, so **no password appears anywhere in this repository**. The username comes from `WRDS_USER` in `.Renvirom`.

Without WRDS access the step fails cleanly and the other six series still download. The primary gold measure is then missing, but the **ASX gold ETF alternative still covers the whole period**, so the analysis can still run.

Verifying a reproduction

No data ships with the repository, so use these to confirm your run matches:

Counts below are as at 1 September 2026. Series that run to the present gain a month with each run; the start dates and `gold_crsp_aud` do not move.

Series	Months	First	Last
<code>as_aus_shares</code> , <code>as_intl_shares</code> , <code>as_fixed_income</code>	217	2008-08	2026-08
<code>art_aus_shares</code>	247	2006-02	2026-08
<code>art_intl_shares</code> , <code>art_bonds</code>	286	2002-11	2026-08
<code>gold_crsp_aud</code>	241	2004-12	2024-12
<code>gold_asx_aud</code>	223	2008-02	2026-08
<code>btc_aud</code>	142	2014-11	2026-08
<code>rf</code>	432	1990-08	2026-07

Also written each run: `data/processed/coverage_report.csv` and `docs/session_info.txt` .

Cross-check on the two gold measures. They are constructed from entirely independent sources — CRSP via WRDS, converted through the RBA exchange rate, versus Yahoo Finance in AUD. Over their 203 overlapping months they correlate **0.948**, with a mean monthly difference of 0.005%. If a reproduction returns a materially lower correlation, something in the collection or conversion has gone wrong. See `data/processed/gold_measure_comparison.csv` .

Two data problems that had to be corrected

Both were found by that cross-check, and both would have corrupted the results silently.

Yahoo timestamps are in exchange-local time, not UTC. A monthly GOLD.AX bar opening on 1 January in Sydney is 13:00 UTC on 31 December. Reading the date in UTC therefore labelled every ASX bar with the *previous month*, shifting the whole series by one month against everything else. `01_collect_data.R` now converts using `meta$exchangeTimezoneName` . BTC-AUD is quoted on UTC and was never affected.

GOLD.AX split 10 for 1, and Yahoo does not adjust for it. `AdjClose` is identical to `Close` , and Yahoo's own split record carries the wrong date (2022-06-08, where the price series shows no break at all). The real discontinuity is December 2010 to January 2011, where the quoted price falls from 134.67 to 13.07 — a fictional -90% month. `02_clean_merge.R` applies the adjustment explicitly and warns if any monthly move above 50% survives.

With both corrected, the two gold proxies agree at 0.948. Before the fixes they correlated -0.001 .

CRSP was checked for the same issue: `cfacpr` is constant at 1 for GLD and no daily move exceeds 15%, so that series needed no adjustment.

Known limitations

The CRSP licence ends 2024-12-31, while every other series runs to 2026-07. This truncates both samples by about 19 months. The proposal's Appendix A1 assumes gold coverage to 2026; in practice the primary gold measure stops at the end of 2024. The ASX gold ETF alternative does cover the full period.

The gold series are ETFs, not spot bullion. Both proxies hold physical gold but drift below the metal over time through management costs. This is a measurement choice, not an error, and should be stated in the write-up.

Yahoo returns monthly bars, not daily. With `range = "max"` the API ignores the `interval` argument. This suits a monthly study, but the code documents it so the behaviour is not mistaken for a daily series.

The current month is excluded. Yahoo returns two rows for the month in progress and its value changes between runs, so `02_clean_merge.R` drops it. This is what makes two runs on different days produce the same panel.

Sources may break without notice. The ART subscription key is embedded in the fund's public JavaScript. The AustralianSuper bot filter could tighten. The RBA parsers assume 10 metadata rows before the header.

ART unit prices carry placeholder zeros before an option began trading. These are removed in cleaning, which is what sets each option's inception date.

Software environment

R version 4.6.0 (2026-04-24 ucrt), Windows 11.

Package	Version		Package	Version
httr2	1.3.0		dplyr	1.2.1
jsonlite	2.0.0		tidyr	1.3.2
readxl	1.5.0		zoo	1.9.0
RPostgres	1.4.10		ggplot2	4.0.3
DBI	1.3.0		scales	1.4.0
dbplyr	2.6.0		here	1.0.2

`docs/session_info.txt` is regenerated by every full run.

Reproducibility practice

- All paths resolve with `here()` from the project root; no absolute paths, no `setwd()`.
- `01_collect_data.R` writes only to `data/raw/`; cleaning is isolated in `02_clean_merge.R`, so raw data is never overwritten.
- Analysis (`03`) is separate from presentation (`04`), so changing a figure never re-runs the analysis or triggers a download.
- Every download step is wrapped so one failing source does not stop the others.
- No credentials in any script. The WRDS username and the ART subscription key come from `.Renviron`, which is gitignored; `.Renviron.example` documents both. The WRDS password never leaves the local `pgpass` file.
- Table and figure file names match the numbering used in the report.

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